

Incident Lifecycle Automation in ServiceNow

PHASE 3: DESIGN

Project Detail	Information
Domain	Information Technology
Project Type	Individual Project
Platform	ServiceNow
Primary Area	IT Service Management (ITSM)
Project Scenario	Corporate VPN incident lifecycle from creation to resolution

Prepared as part of the seven-phase project documentation

1. Phase Overview

The design phase translates the requirements into a clear ServiceNow process model. The goal is to decide how the service hierarchy, incident record, support roles, knowledge, change records, child incidents, and SLA information should fit together before the implementation is demonstrated. The design intentionally uses standard ITSM concepts so that the project is easy to explain and maintain.

2. High-Level Solution Architecture

The project is designed around the Incident record as the central coordination point. The affected service and service offering provide business context, the configuration item provides technical context, Knowledge Management supports troubleshooting, the assignment group controls ownership, Change Management supports controlled technical modifications, child incidents represent related users or occurrences, and SLA records provide timing visibility.

Layer / component	Design role
Service layer	Remote Access and Corporate VPN identify what business capability is affected.
Incident Management	Acts as the primary workflow record from issue reporting through resolution.
Knowledge Management	Provides troubleshooting guidance and stores the final reusable solution.
Assignment / support groups	Separates first-line Service Desk handling from Network / Level 2 technical investigation.
Change Management	Controls the emergency change required when the resolution changes the environment.
Related Incident model	Links a child incident when the same issue affects another user or related case.
SLA / related lists	Provides timing and relationship evidence during final validation.

3. Service Hierarchy Design

The service hierarchy gives the incident meaningful business context. Instead of recording only a generic network problem, the incident is associated with the Remote Access service and the Corporate VPN service offering. This makes the scenario easier to understand because it clearly answers what service is affected and which specific offering the employee is trying to use.

Level	Designed value	Reason
Business Service	Remote Access	Represents the broader ability for employees to access corporate resources remotely.
Service Offering	Corporate VPN	Represents the specific remote-access offering used in the incident scenario.
Configuration Item	Available CI used in the demonstration, where applicable	Adds technical context to support investigation and change activity.

4. Incident Process Flow Design

1. User reports VPN issue
2. Incident is created

3. Incident is classified and linked to service context
4. Knowledge / Agent Assist is checked
5. Service Desk performs initial handling
6. Incident is reassigned to Network / Level 2 when needed
7. Level 2 records investigation details
8. Emergency change is created and linked if a change is required
9. Child incident is linked if another user is affected
10. Cause and resolution are documented
11. Parent and related incidents are resolved as appropriate
12. Knowledge article is created from the verified solution
13. SLA and related records are validated

5. Incident State and Ownership Design

The design separates the incident state from the assignment owner. A record can remain active while responsibility moves from the Service Desk to the Network team. This is important because reassignment should not break the history of the incident. The activity stream and work notes should show why the handoff occurred and what information was available to the receiving team.

Stage	Typical owner	Design intention
New / initial review	Service Desk	Capture the problem, verify basic information, and classify the incident.
Initial troubleshooting	Service Desk	Use available knowledge and perform first-line checks.
Escalated investigation	Network / Level 2	Perform deeper technical analysis and determine whether a change is required.
Change activity	Network / Change participants	Execute or track the controlled technical fix.
Resolution and validation	Owning support team	Record the solution, confirm relationships, and move the incident to resolved.

6. Knowledge Management Design

Knowledge is used twice in the lifecycle. At the beginning, existing knowledge can help the Service Desk avoid unnecessary escalation when a known solution already exists. At the end, the final verified solution can be converted into a new or improved knowledge article. This creates a feedback loop: each resolved incident can make the next similar incident easier to solve.

- Search for relevant VPN troubleshooting information during initial analysis.
- Use Agent Assist or the available knowledge search experience where supported by the workspace.
- Avoid treating an article as correct simply because it appears in search results; the support user should confirm that it matches the incident symptoms.
- After resolution, capture the verified cause, steps, and outcome in a reusable article using clear language.

7. Change Management Design

The change record is not created for every incident. It is introduced only when the solution requires a controlled modification to the environment. In the project scenario, this is represented by an emergency change because the incident affects active VPN access and the technical team may need to make an urgent configuration or infrastructure correction. The key design requirement is traceability: the evaluator should be able to see that the change exists because of the incident and that the incident shows the related change.

8. Parent and Child Incident Design

A child incident is used to demonstrate how ServiceNow can relate another affected user or related occurrence to the same underlying issue. The parent record carries the main investigation, while the child record provides a separate user-facing incident that remains connected to the central problem. This design reduces duplicated investigation and helps explain how multiple reports can be managed together.

9. Data Relationship Design

Source record	Relationship	Target record
Incident	Affected by / associated with	Remote Access service
Incident	Uses / affected offering	Corporate VPN service offering
Incident	May reference	Configuration Item
Incident	Assigned to	Service Desk or Network group
Incident	Supported by	Knowledge article
Incident	Related to	Emergency change
Parent Incident	Has child	Child Incident
Incident	Measured by	Task SLA / SLA information

10. User Experience Design

The project should be demonstrated through a workspace or standard form experience that makes the main fields and related records easy to see. The design avoids unnecessary customization because the objective is to prove the lifecycle rather than to build a completely new user interface. The most important information should be visible in a logical sequence: caller and short description, service context, classification, assignment, activity, related records, and resolution details.

11. Access and Security Considerations

ServiceNow uses roles and access controls to determine what users can see and modify. In a training environment, the project may use an administrator or elevated user for configuration and may impersonate support users during the demonstration. Even in an academic project, good practice is to avoid sharing personal passwords, instance credentials, or real employee data in screenshots, recordings, GitHub repositories, or documentation.

12. Exception and Failure Design

A good demonstration design also considers what happens when the expected path is not available. For example, Agent Assist may not return a useful article, the desired configuration item may not exist, or a

user may not have permission to create a change. The project should handle these situations transparently: use an available sample record where appropriate, explain the limitation, and avoid pretending that a feature worked when it did not.

13. Design Validation Checklist

- The service hierarchy clearly represents Remote Access and Corporate VPN.
- The incident is the central record throughout the lifecycle.
- Assignment can move from Service Desk to Network / Level 2 without losing history.
- Knowledge is present both as troubleshooting support and as a post-resolution output.
- Change Management is used only when a controlled technical change is needed.
- The child incident remains visibly related to the parent incident.
- SLA information and related records can be reviewed during the final validation.

14. Phase Conclusion

The design phase provides the blueprint for the implementation. It shows how the ServiceNow records relate to one another, how ownership changes during the lifecycle, and how knowledge, change, child incidents, and SLA information support the incident record. With this design in place, the development work can follow a clear sequence instead of being performed as isolated configuration tasks.